Third storyboard: "Always two, over $\mathbb{C}$"

Reply Short to Inigo Quilez: `https://www.youtube.com/shorts/cc-Z_igk4Lo`. A third, independent story (story first). v4 and alternative v2 are unchanged.

Format. YouTube Short, 9:16 (1080×1920). **178 s**, 30 scenes, 469 words of voiceover (≈2.6 words/s). No audio or MP4 at this stage.
Look. Same chalkboard as v4 and alt v2. Mirrors chalk-white, circles **blue** (dashed = image), **orange = imaginary**, grey = a pencil of circles. In the 3D plots colour blends blue→orange with $|\mathrm{Im}\, y|/(|\mathrm{Re}\, y| + |\mathrm{Im}\, y|)$.
Opening. First person (Carlo): "Inigo Quilez's video is lovely, with many directions to explore. I chose one…", with the credit caption on screen.
Through-line. A circle is four complex numbers $[a : b : c : d] \in \mathbb{P}^3(\mathbb{C})$. Reflecting a circle in a circle, $C \star L = C - \frac{2\langle C,L\rangle}{Q(L)} L$, is closed (a kei; the mirrors generate the complex Möbius group $PO(4,\mathbb{C})$). $C \star L$ meets C exactly where L does, so "where a circle meets a line" is a question inside the algebra: always two points (with I and J, four). One invariant, $\kappa = \langle C, L\rangle^2/(Q(C)Q(L))$, decides tangency ($\kappa = 1$). Realness is not invariant; "hidden" is what a real slice does to one symmetric picture.

Guardrails. (1) *Convention.* Our line $x = t$ slides perpendicular to itself and we place the points at its foot: $(t, \pm\sqrt{1-t^2})$, $(t, \pm is)$, $t^2 - s^2 = 1$. Quilez's line translates and rotates and his orange dots sit at $\mathbf{o} \pm \sqrt{h^2 - r^2}\,\mathbf{d}$, centred on his moving point $\mathbf{o}$; his second formula only inserts an orange i (radicand unchanged). We do not reproduce his picture (scenes 3, 5 say so). (2) $C \star L$ needs $Q(L) \neq 0$ (no point-circles or isotropic lines as mirrors); $C \cap (C \star L) = C \cap L$ needs $\langle C, L\rangle \neq 0$ (else $C \star L = C$). (3) "Always two": with multiplicity, $a \neq 0$, $Q(L) \neq 0$; an isotropic line meets a circle in one finite point. (4) The invariant is named κ because I, J denote the circular points. (5) $PO(4,\mathbb{C})$ generation is Cartan–Dieudonné (cited, not computed); κ invariance and equivariance are checked numerically. (6) A complex circle's real points are $C_1 \cap C_2$; two/one/none by the sign of $\Delta_{\mathbb{R}}$, or a whole circle if C is real up to scale. (7) For $x^2 + y^2 + i\varepsilon x = 1$ the two points never merge for real t (tangency at $t = \pm 0.989 - 0.15i$) and are both real only at $t = 0$.

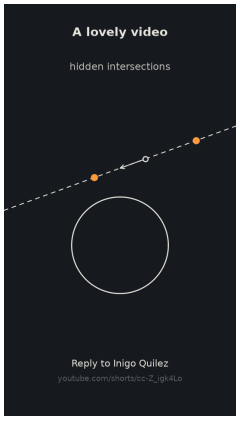
3D choice. Used: $(t, \mathrm{Re}\, y, \mathrm{Im}\, y)$ for the sweep (scenes 8, 24–27): the circle lies in the plane $\mathrm{Im}\, y = 0$, the hyperbola in $\mathrm{Re}\, y = 0$, and they join at $t = \pm 1$; nudged by ε the joints visibly come apart, which 2D cannot show. Not used: points of $\mathbb{C}^2$ as three real coordinates plus colour for the fourth (hard to read in a still or a 5 s beat), and colour = κ (along the real sweep $\kappa = t^2$ is just a function of the t axis, so it adds nothing; scene 20 shows it as a 2D parabola instead).

Files: frames `alt3_frames/` by `alt3_frames.py`, numbers in `alt3_frames/check_log.txt`; claims checked by `alt3_verify.py` (`alt3_verify_log.txt`, all PASS); review passes in `alt3_review_log.txt`.

act	scenes	time
Opening (first person, credit)	1–2	0:00–0:10
His move, our slice, one curve (3D)	3–8	0:10–0:47
Circles in $\mathbb{P}^3(\mathbb{C})$: the operation, a kei	9–14	0:47–1:25
Always two; four, really	15–17	1:25–1:48
The invariant κ, tangency, not realness	18–21	1:48–2:12
A complex circle's real trace	22–23	2:12–2:25
Nudge off the reals (3D)	24–27	2:25–2:45
Recap and thanks	28–30	2:45–2:58

#	time	beat	voiceover
1	0:00–0:04	A lovely video	*Inigo Quilez's video is lovely, with many directions to explore.*
2	0:04–0:10	The one I chose	*I chose one: circles acting on circles, with complex numbers everywhere. One closed operation, one invariant.*
3	0:10–0:15	His move	*His move: when the square root goes negative, the intersection points turn complex.*
4	0:15–0:21	Our slice	*Our slice: a unit circle and a vertical line, x equals t, sliding without turning.*
5	0:21–0:28	Not his picture	*His orange dots hang on his moving point o; ours sit at the foot of the perpendicular. Our version, not his.*
6	0:28–0:34	Real, then hidden	*Inside, the points are t, plus or minus a real root. Outside, they're plus or minus i s.*
7	0:34–0:40	A hyperbola	*Drop the i, plot s, and you get a hyperbola: t squared minus s squared equals one.*
8	0:40–0:47	One curve	*Better: give the imaginary part its own axis. Circle and hyperbola join into one curve, meeting where the line touches.*
9	0:47–0:54	Four complex numbers	*A circle is four numbers: a, b, c, d. Let them be complex; only ratios matter, and lines have a zero.*
10	0:54–1:00	Its Q	*Each circle carries one number, Q. For a real circle it's four times the radius squared.*
11	1:00–1:05	The operation	*The operation: reflect C in L by subtracting just the right multiple of L.*
12	1:05–1:12	Familiar, and closed	*For real circles, that's reflection in a line, or inversion in a circle. Circle in, circle out, even over C.*
13	1:12–1:18	One exception	*The only forbidden mirrors have Q equal to zero: point-circles, and lines through the circular points.*
14	1:18–1:25	Three rules: a kei	*The rules: a mirror fixes itself, reflecting twice brings you back, and mirrors carry relations along. That makes a kei.*
15	1:25–1:33	Always two	*Where does circle C meet line L? Substituting gives a quadratic with leading term a times Q of L: always two points.*
16	1:33–1:40	Through the mirror	*Reflect C in L: C star L is C minus a multiple of L, so it meets C where L does.*

17	1:40–1:48	Four, really	*As curves, two circles meet four times: twice at the circular points I and J, shared by every circle, and twice on our line.*
18	1:48–1:55	The invariant	*Chain mirrors and you get the complex Möbius group. One number survives all of it: kappa.*
19	1:55–2:00	Tangency	*Kappa equals one exactly when the two points merge: tangency.*
20	2:00–2:06	In our slice	*In our slice, kappa is just t squared: below one they cross, above one they hide.*
21	2:06–2:12	Not realness	*Complex maps keep the two points, tangency and kappa, but not whether the points are real.*
22	2:12–2:19	A complex circle	*A complex circle splits into two real circles: its real and imaginary parts. Its real points are where they cross.*
23	2:19–2:25	Two or none	*So it has two real points, one, or none, and a similar test decides.*
24	2:25–2:30	Nudge the circle	*Now nudge our unit circle off the reals, and sweep the line again.*
25	2:30–2:36	They come apart	*The junctions come apart: tangency now happens at complex t, so the real sweep never splits.*
26	2:36–2:40	Two, all the way	*Two points all the way. No switch, no branching.*
27	2:40–2:45	Hidden is a slice	*Hidden is just what a real slice does to one symmetric picture.*
28	2:45–2:50	Recap	*One operation, circle star circle. One invariant, kappa, with tangency at one.*
29	2:50–2:54	Always two, over C	*Always two points, real or not.*
30	2:54–2:58	Thanks	*Thanks, Inigo, for the spark.*

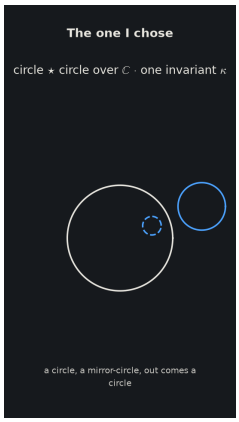


Scene 1: A lovely video 0:00–0:04 (4 s)

VO: *Inigo Quilez's video is lovely, with many directions to explore.*

On screen: "Reply to Inigo Quilez — youtube.com/shorts/cc-Z_igk4Lo"

Animation: His setup, schematic: a circle, a tilted line that misses it, two orange hidden dots. Credit caption fades in. First person, Carlo.

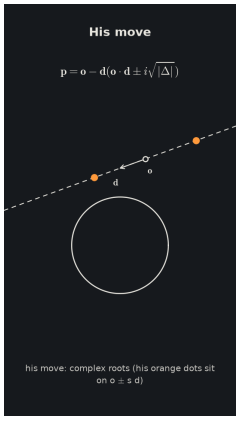


Scene 2: The one I chose 0:04–0:10 (6 s)

VO: *I chose one: circles acting on circles, with complex numbers everywhere. One closed operation, one invariant.*

On screen: circle ⋆ circle over $\mathbb{C}$ · one invariant κ

Animation: A mirror circle flips a blue circle into its dashed image.

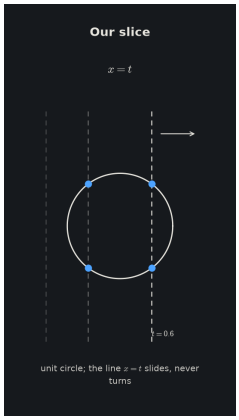


Scene 3: His move 0:10–0:15 (5 s)

VO: *His move: when the square root goes negative, the intersection points turn complex.*

On screen: $\mathbf{p} = \mathbf{o} - \mathbf{d}\big(\mathbf{o}\cdot\mathbf{d} \pm i\sqrt{|\Delta|}\big)$

Animation: His formula, rewritten with $\sqrt{|\Delta|}$ (on his screen the orange i is inserted and the radicand left unchanged); orange dots on his line, centred on his point $\mathbf{o}$.

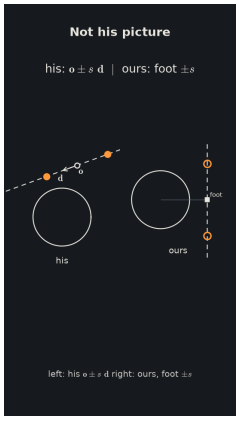


Scene 4: Our slice 0:15–0:21 (6 s)

VO: *Our slice: a unit circle and a vertical line, x equals t, sliding without turning.*

On screen: $x = t$

Animation: The vertical line slides left to right; two blue dots ride the circle.

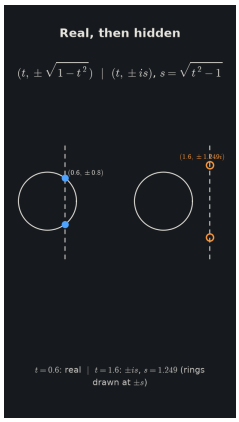


Scene 5: Not his picture 0:21–0:28 (7 s)

VO: *His orange dots hang on his moving point o; ours sit at the foot of the perpendicular. Our version, not his.*

On screen: his: $\mathbf{o} \pm s\,\mathbf{d}$ | ours: foot $\pm s$

Animation: Split screen: left his convention, right ours.

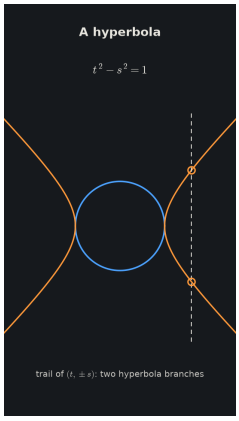


Scene 6: Real, then hidden 0:28–0:34 (6 s)

VO: *Inside, the points are t, plus or minus a real root. Outside, they're plus or minus i s.*

On screen: $(t, \pm\sqrt{1-t^2})$ | $(t, \pm is)$, $s = \sqrt{t^2-1}$

Animation: Left $t = 0.6$: blue dots. Right $t = 1.6$: orange rings, labelled $\pm is$.

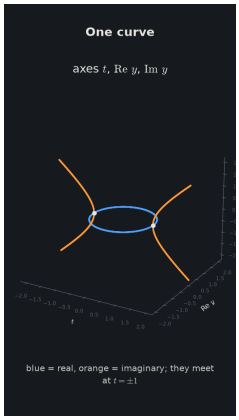


Scene 7: A hyperbola 0:34–0:40 (6 s)

VO: *Drop the i, plot s, and you get a hyperbola: t squared minus s squared equals one.*

On screen: $t^2 - s^2 = 1$

Animation: The orange rings leave a trail: two hyperbola branches beside the blue circle.



Scene 8: One curve 0:40–0:47 (7 s)

VO: *Better: give the imaginary part its own axis. Circle and hyperbola join into one curve, meeting where the line touches.*

On screen: axes t, $\mathrm{Re}\, y$, $\mathrm{Im}\, y$

Animation: 3D: all points (t, y) of $x^2 + y^2 = 1$ with $x = t$ real. Then y^2 is real, so y is real or purely imaginary: the circle (plane $\mathrm{Im}\, y = 0$) plus the hyperbola (plane $\mathrm{Re}\, y = 0$), meeting at $t = \pm 1$. Colour: blue = real, orange = imaginary. Slow orbit.

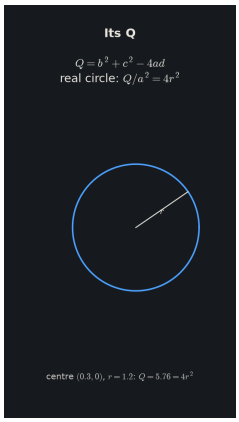


Scene 9: Four complex numbers 0:47–0:54 (7 s)

VO: *A circle is four numbers: a, b, c, d. Let them be complex; only ratios matter, and lines have a zero.*

On screen: $a(x^2+y^2)+bx+cy+d=0$, $[a:b:c:d]\in\mathbb{P}^3(\mathbb{C})$; lines: $a=0$

Animation: Three real circles and a line, each tagged with its four numbers; the tags turn complex and a common factor λ is crossed out.

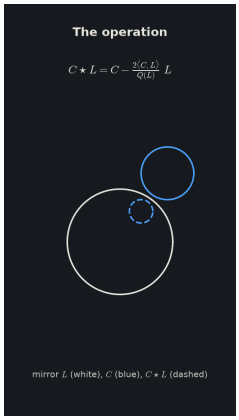


Scene 10: Its Q 0:54–1:00 (6 s)

VO: *Each circle carries one number, Q. For a real circle it's four times the radius squared.*

On screen: $Q=b^2+c^2-4ad$; real circle: $Q/a^2=4r^2$

Animation: Radius drawn; Q computed.

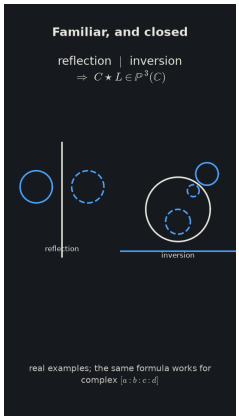


Scene 11: The operation 1:00–1:05 (5 s)

VO: *The operation: reflect C in L by subtracting just the right multiple of L.*

On screen: $C\star L=C-\frac{2\langle C,L\rangle}{Q(L)}L$

Animation: Formula slams in over a mirror circle L and blue C.

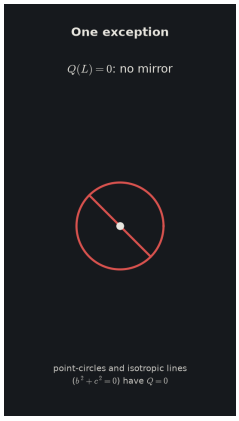


Scene 12: Familiar, and closed 1:05–1:12 (7 s)

VO: *For real circles, that's reflection in a line, or inversion in a circle. Circle in, circle out, even over C.*

On screen: line: reflection | circle: inversion $\Rightarrow\ C\star L\in\mathbb{P}^3(\mathbb{C})$

Animation: Two panels: a circle reflected in a line; a circle and a line inverted in a circle.

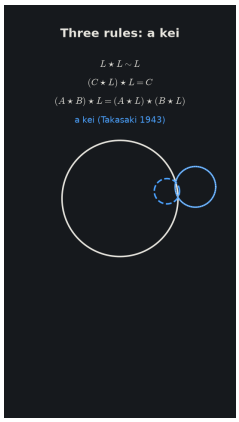


Scene 13: One exception 1:12–1:18 (6 s)

VO: *The only forbidden mirrors have Q equal to zero: point-circles, and lines through the circular points.*

On screen: $Q(L) = 0$: no mirror (point-circles, centre anywhere in $\mathbb{C}^2$ or at ∞; lines with $b^2 + c^2 = 0$)

Animation: A point-circle with a no-entry mark; text: lines with $b^2 + c^2 = 0$.

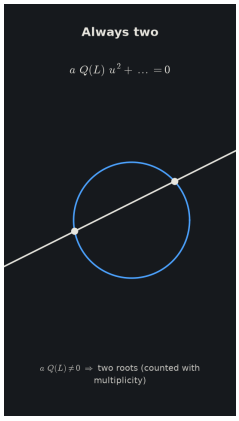


Scene 14: Three rules: a kei 1:18–1:25 (7 s)

VO: *The rules: a mirror fixes itself, reflecting twice brings you back, and mirrors carry relations along. That makes a kei.*

On screen: $L \star L \sim L$ (it is $-L$); $(C \star L) \star L = C$; $(A \star B) \star L = (A \star L) \star (B \star L)$; a kei (Takasaki 1943)

Animation: Rules type on one per beat; the word kei lands last.

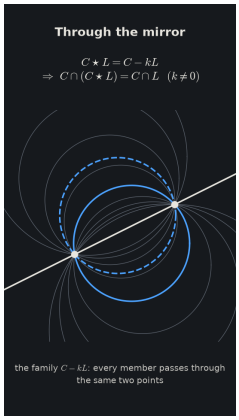


Scene 15: Always two 1:25–1:33 (8 s)

VO: *Where does circle C meet line L? Substituting gives a quadratic with leading term a times Q of L: always two points.*

On screen: $a\,Q(L)\,u^2 + \cdots = 0$

Animation: A circle and a crossing line; the quadratic writes on; two blue dots.

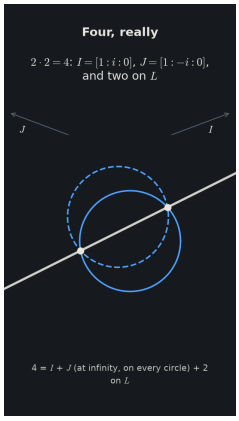


Scene 16: Through the mirror 1:33–1:40 (7 s)

VO: *Reflect C in L: C star L is C minus a multiple of L, so it meets C where L does.*

On screen: $C \star L = C - kL \Rightarrow C \cap (C \star L) = C \cap L$ (if $k \neq 0$; $k = 0$ means $C \star L = C$)

Animation: $C \star L$ appears dashed through the same two dots; the family $C - kL$ fans out through them.

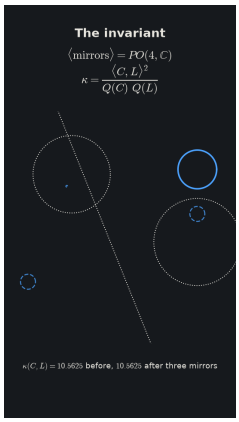


Scene 17: Four, really 1:40–1:48 (8 s)

VO: *As curves, two circles meet four times: twice at the circular points I and J, shared by every circle, and twice on our line.*

On screen: $2 \cdot 2 = 4$: $I = [1 : i : 0]$, $J = [1 : -i : 0]$, and two on L

Animation: Bézout count; two arrows off to infinity labelled I, J; then L glows between C and $C \star L$.

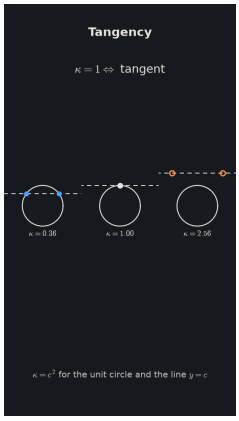


Scene 18: The invariant 1:48–1:55 (7 s)

VO: *Chain mirrors and you get the complex Möbius group. One number survives all of it: kappa.*

On screen: $\langle \text{mirrors} \rangle = PO(4, \mathbb{C})$; $\kappa = \frac{\langle C,L \rangle^2}{Q(C)Q(L)}$

Animation: A circle bounces through three mirrors; then the formula for κ, with the same value on both configurations.

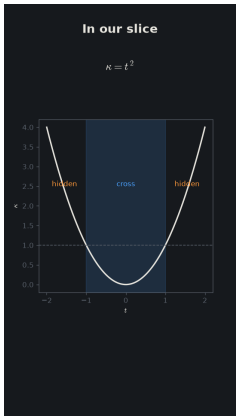


Scene 19: Tangency 1:55–2:00 (5 s)

VO: *Kappa equals one exactly when the two points merge: tangency.*

On screen: $\kappa = 1 \iff$ tangent

Animation: Three panels $\kappa < 1$, $= 1$, > 1.

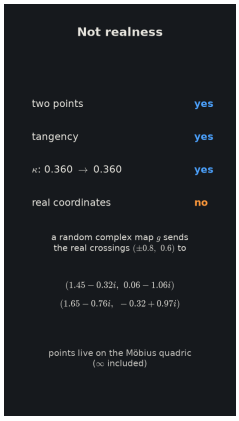


Scene 20: In our slice 2:00–2:06 (6 s)

VO: *In our slice, kappa is just t squared: below one they cross, above one they hide.*

On screen: $\kappa = t^2$

Animation: Parabola $\kappa(t)$ with the crossing band.

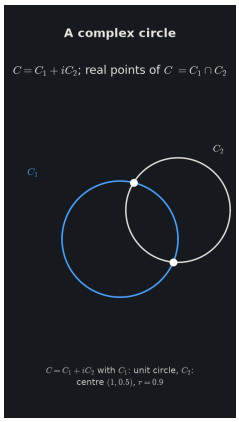


Scene 21: Not realness 2:06–2:12 (6 s)

VO: *Complex maps keep the two points, tangency and kappa, but not whether the points are real.*

On screen: random $g \in PO(4, \mathbb{C})$: κ 0.360 → 0.360; still two points, now with complex coordinates (points on the Möbius quadric, ∞ included)

Animation: Checklist.

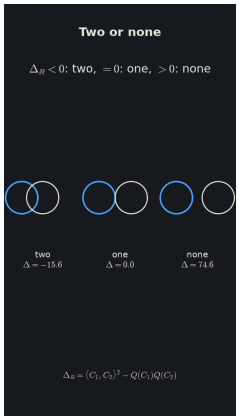


Scene 22: A complex circle 2:12–2:19 (7 s)

VO: *A complex circle splits into two real circles: its real and imaginary parts. Its real points are where they cross.*

On screen: $C = C_1 + iC_2$; real points of $C = C_1 \cap C_2$

Animation: C_1 and C_2 drawn; their crossings glow.

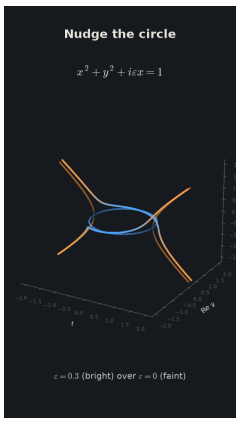


Scene 23: Two or none 2:19–2:25 (6 s)

VO: *So it has two real points, one, or none, and a similar test decides.*

On screen: $\langle C_1, C_2 \rangle^2 - Q(C_1)Q(C_2)$: < 0 two, $= 0$ one, > 0 none (whole circle if C is real up to scale)

Animation: Three panels.

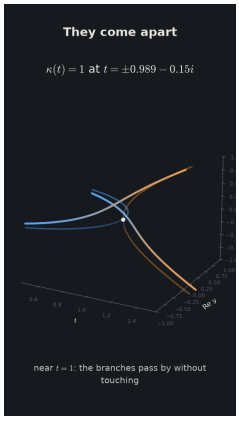


Scene 24: Nudge the circle 2:25–2:30 (5 s)

VO: *Now nudge our unit circle off the reals, and sweep the line again.*

On screen: $x^2 + y^2 + i\varepsilon x = 1$

Animation: 3D curves for $\varepsilon = 0.3$, ghost of $\varepsilon = 0$.

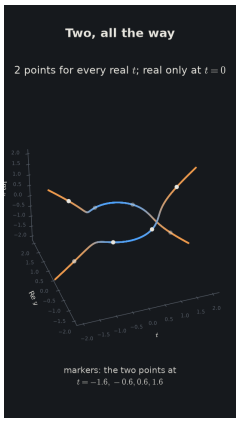


Scene 25: They come apart 2:30–2:36 (6 s)

VO: *The junctions come apart: tangency now happens at complex t, so the real sweep never splits.*

On screen: $\kappa(t) = 1$ at $t = \pm 0.989 - 0.15i$

Animation: Zoom on the gap near $t = 1$.

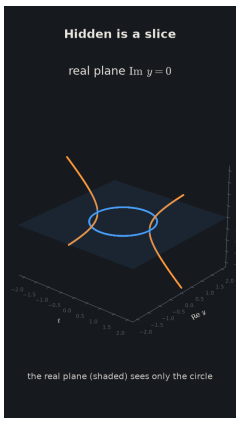


Scene 26: Two, all the way 2:36–2:40 (4 s)

VO: *Two points all the way. No switch, no branching.*

On screen: 2 points for every real t; real only at $t = 0$

Animation: 3D orbit.

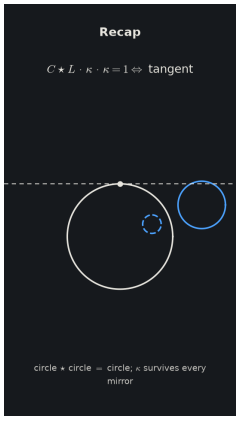


Scene 27: Hidden is a slice 2:40–2:45 (5 s)

VO: *Hidden is just what a real slice does to one symmetric picture.*

On screen: real plane $\mathrm{Im}\, y = 0$

Animation: Back to $\varepsilon = 0$; the real plane is shaded.

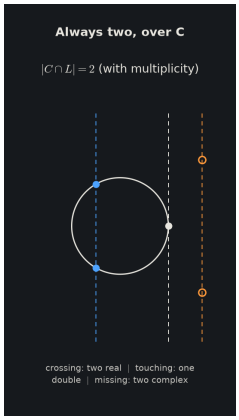


Scene 28: Recap 2:45–2:50 (5 s)

VO: *One operation, circle star circle. One invariant, kappa, with tangency at one.*

On screen: $C \star L \cdot \kappa \cdot \kappa = 1 \iff$ tangent

Animation: Recap.

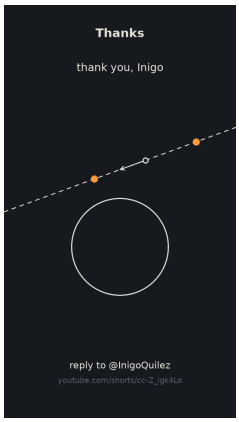


Scene 29: Always two, over C 2:50–2:54 (4 s)

VO: *Always two points, real or not.*

On screen: $\#(C \cap L) = 2$ (with multiplicity; $a \neq 0$, $Q(L) \neq 0$)

Animation: Recap.



Scene 30: Thanks 2:54–2:58 (4 s)

VO: *Thanks, Inigo, for the spark.*

On screen: reply to @InigoQuilez

Animation: Final hold.

Full voiceover script (178 s, 469 words)

1. [0:00–0:04] Inigo Quilez's video is lovely, with many directions to explore.
2. [0:04–0:10] I chose one: circles acting on circles, with complex numbers everywhere. One closed operation, one invariant.
3. [0:10–0:15] His move: when the square root goes negative, the intersection points turn complex.
4. [0:15–0:21] Our slice: a unit circle and a vertical line, x equals t, sliding without turning.
5. [0:21–0:28] His orange dots hang on his moving point o; ours sit at the foot of the perpendicular. Our version, not his.
6. [0:28–0:34] Inside, the points are t, plus or minus a real root. Outside, they're plus or minus i s.
7. [0:34–0:40] Drop the i, plot s, and you get a hyperbola: t squared minus s squared equals one.
8. [0:40–0:47] Better: give the imaginary part its own axis. Circle and hyperbola join into one curve, meeting where the line touches.
9. [0:47–0:54] A circle is four numbers: a, b, c, d. Let them be complex; only ratios matter, and lines have a zero.
10. [0:54–1:00] Each circle carries one number, Q. For a real circle it's four times the radius squared.
11. [1:00–1:05] The operation: reflect C in L by subtracting just the right multiple of L.
12. [1:05–1:12] For real circles, that's reflection in a line, or inversion in a circle. Circle in, circle out, even over C.
13. [1:12–1:18] The only forbidden mirrors have Q equal to zero: point-circles, and lines through the circular points.
14. [1:18–1:25] The rules: a mirror fixes itself, reflecting twice brings you back, and mirrors carry relations along. That makes a kei.
15. [1:25–1:33] Where does circle C meet line L? Substituting gives a quadratic with leading term a times Q of L: always two points.
16. [1:33–1:40] Reflect C in L: C star L is C minus a multiple of L, so it meets C where L does.
17. [1:40–1:48] As curves, two circles meet four times: twice at the circular points I and J, shared by every circle, and twice on our line.
18. [1:48–1:55] Chain mirrors and you get the complex Möbius group. One number survives all of it: kappa.
19. [1:55–2:00] Kappa equals one exactly when the two points merge: tangency.
20. [2:00–2:06] In our slice, kappa is just t squared: below one they cross, above one they hide.
21. [2:06–2:12] Complex maps keep the two points, tangency and kappa, but not whether the points are real.
22. [2:12–2:19] A complex circle splits into two real circles: its real and imaginary parts. Its real points are where they cross.
23. [2:19–2:25] So it has two real points, one, or none, and a similar test decides.
24. [2:25–2:30] Now nudge our unit circle off the reals, and sweep the line again.
25. [2:30–2:36] The junctions come apart: tangency now happens at complex t, so the real sweep never splits.
26. [2:36–2:40] Two points all the way. No switch, no branching.
27. [2:40–2:45] Hidden is just what a real slice does to one symmetric picture.
28. [2:45–2:50] One operation, circle star circle. One invariant, kappa, with tangency at one.
29. [2:50–2:54] Always two points, real or not.
30. [2:54–2:58] Thanks, Inigo, for the spark.